

COURSE TITLE : STATISTICAL QUALITY ASSURANCE IN TEXTILES
COURSE CODE : 5064
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Statistical methods	13
2	Probability and significance test	13
3	Quality control charts	13
4	Management and organization of Quality control	13
TOTAL		52

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand the construction frequency distribution , graphical representation of frequency distribution and different types of frequency curves
	2	Know the measures of central tendency and various measures of dispersion
2	1	Understand the clear idea about probability.
	2	Know the significance tests on Mean & SD
3	1	Understand the construction of various quality control charts
4	1	Know the various management and organization of quality control

SPECIFIC OUTCOME

MODULE I STATISTICAL METHODS

1.1.0 Understand the Construction Frequency distribution, Graphical representation of frequency distribution and Different types of frequency curves

- 1.1.1** Differentiate between Quality control & Quality Assurance
- 1.1.2** State the terms population and individual in statistics
- 1.1.3** Classify the tabulation of data
- 1.1.4** Explain the preparation of frequency table
- 1.1.5** Draw the Different types of Graphical representation of frequency distribution
- 1.1.6** Draw the Different types of frequency curves

1.2.0 Know the Measures of central Tendency and various Measures of Dispersion

- 1.2.1** List the Measures of central Tendency.
- 1.2.2** Explain the various Measures of Dispersion
- 1.2.3** State the relationship between mean range, mean deviation and S D
- 1.2.4** List the Characteristics of normal distribution curve

MODULE II PROBABILITY AND SIGNIFICANCE TEST

2.1.0 Understand the clear idea about Probability

- 2.1.1 Define the Probability theory
- 2.1.2 Discuss the Classical determination of probability
- 2.1.3 Differentiate between Sampling and hundred percent inspections
- 2.1.4 State the Standard error of mean and Standard error of SD

2.2.0 Know the Significance tests on Mean & SD

- 2.2.1 State the types of Significance Test
- 2.2.2 Explain the Significance test on Single mean of small sample and large sample
- 2.2.3 Describe the Significance test on means of small sample and means of large sample
- 2.2.4 Explain the Significance test on SD

MODULE III QUALITY CONTROL CHARTS

3.1.0 Understand the construction of various Quality Control charts

- 3.1.1 State the Meaning of control charts
- 3.1.2 List the types of quality control charts
- 3.1.3 State the uses of quality control charts
- 3.1.4 Describe the construction of X and R chart
- 3.1.5 Determination of control limits
- 3.1.6 Illustrate the interpretation of control chart

MODULE IV MANAGEMENT AND ORGANIZATION OF QUALITY CONTROL

4.1.0 Know the various Management and organization of Quality control

- 4.1.1 State the Management techniques for quality control
- 4.1.2 Describe the statistical concepts and its tolerances.
- 4.1.3 Explain the machine interference with special reference to textile industry
- 4.1.4 Describe the Snap survey techniques in spinning & weaving industry.
- 4.1.5 List the specification of quality standards in Spinning, weaving & garment industry.
- 4.1.6 State the application and maintenance of SQC systems adopted in Textile mills.
- 4.1.7 Define the total Quality control.
- 4.1.8 Describe Planning and organization of Total Quality Control in mills
- 4.1.9 Determine the cost reduction and quality control

CONTENT DETAILS

MODULE I STATISTICAL METHODS

Definition of statistics - Difference between Quality control & Quality Assurance -Meaning of population and individual in statistics - Classification and tabulation of data –Preparation of continuous frequency distribution - Different types of Graphical representation of frequency distribution-Histogram & Frequency polygon - Different types of frequency curves - Normal distribution curve, Asymmetric distribution curve, Bi-modal distribution curve -Measures of central Tendency – Mean, Median, Mode -

Measures of Dispersion - Range, Mean range, Mean deviation, Percentage mean deviation, Standard deviation & Co-efficient of variation – Characteristics of normal distribution curve - Relationship between mean range, mean deviation and SD.

MODULE II PROBABILITY AND SIGNIFICANCE TEST

Probability theory - Classical determination of probability - Sampling versus hundred percent inspections - Standard error - Single mean of small sample, Single mean of large sample, Between means of small sample, Between means of large sample - Significance Test - Significance test on Mean -Significance test on SD - Single SD of small sample, Single SD of large sample, Between SD's of small sample, Between SD's of large sample

MODULE III QUALITY CONTROL CHARTS

Concept of quality control charts- Meaning of control charts -Uses of quality control charts -Types of quality control charts - X and R chart - Fraction defective chart (p Chart), Number defective chart (c Chart) - Determination of control limits - Construction of control chart - Interpretation of control chart

MODULE IV MANAGEMENT AND ORGANIZATION OF QUALITY CONTROL

Management techniques for quality control - Statistical concepts and its tolerances –Machine interference with special reference to textile industry - Snap survey techniques -Specification of quality standards in Textile Mills - Application and maintenance of SQC systems adopted in Textile mills - Total Quality control - Planning and organization of Total Quality Control in mills - Cost reduction and quality control.

REFERENCE BOOKS

S.No	Title Author	Name of book	Publishers
1.	J. E. Booth	Principles of Textile testing	Mahajan publishers Ahammedbad
2.	B.Grover and D.S.Hamby	Hand Book of Textile Test and QC	
3.	T. Murphy, Tipper & KM. Morris	Statistical Methods for Textile Technologists	
4.	S.P. Gupta	Statistical Methods	Universal Publishing corp.,Mumbai -2
5.	SITRA Publication	Methods of Statistics	Universal Publishing corp.,Mumbai -2